ESP Lab1: Short Report

Signal Modulation by Decimation and Interpolation

Group / Team	**Team member 1:**	Pham, Tuong Minh
ESPL01 / 5 (please change)	**Team member 2:**	Ul Haq, Waleed
	Team member 3:	First, Last name (please change)

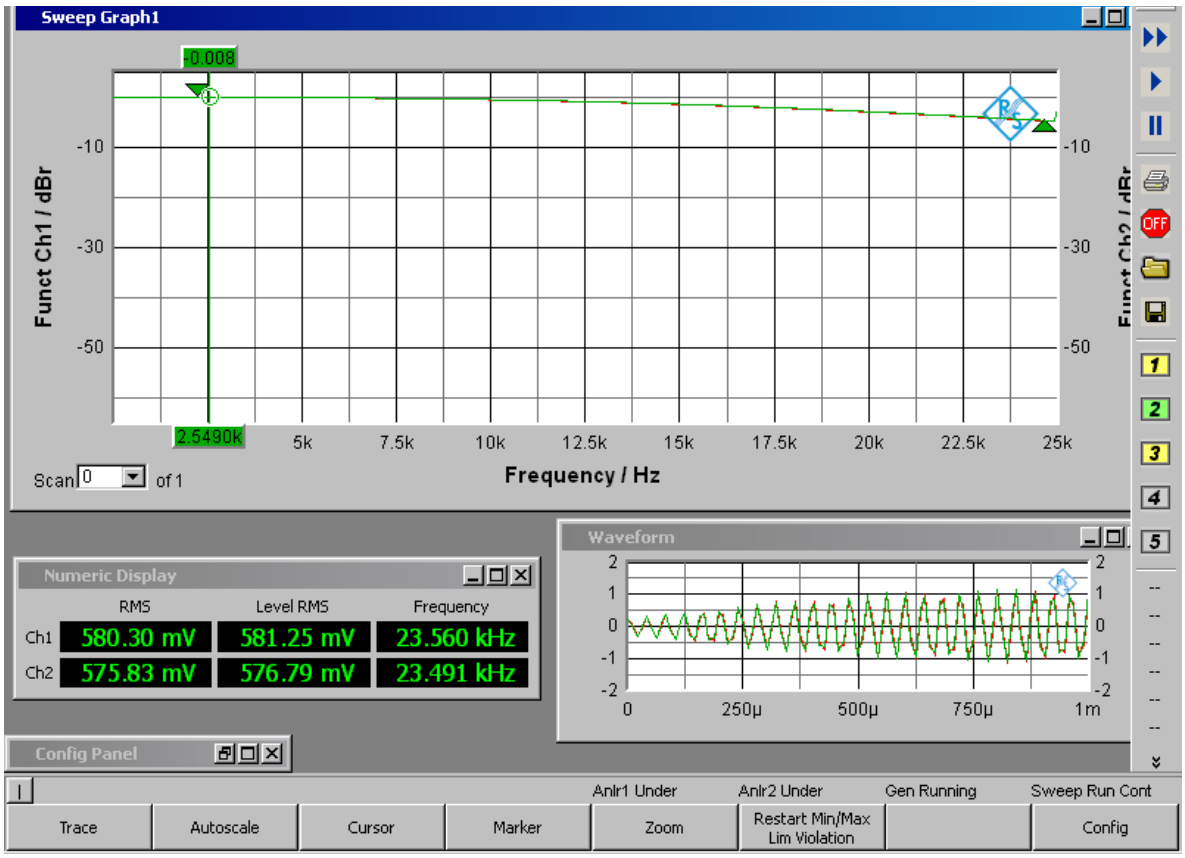
Table 1: Different configurations to be used by different lab groups

Lab group as chosen in myHAW	**Down- / Upsampling factor**	**Chirp, from ... to [kHz], length [s]**	**Bat sound** f_{bat}	**Sampling frequency ADC, DAC** f_s [kHz]	**Polyphases** x_λ **output at DAC CH0 / CH1**
ESPL/01	M=L=3	18 ... 21 kHz 2.5 s	18 kHz	50 kHz, 50 kHz	$\lambda = 0$ / $\lambda = 2$
ESPL/02	M=L=4	14 ... 17 kHz 3.0 s	15 kHz	50 kHz, 50 kHz	$\lambda = 0$ / $\lambda = 3$
ESPL/03	M=L=5	21 ... 24 kHz 2.0 s	17 kHz	50 kHz, 50 kHz	$\lambda = 0$ / $\lambda = 4$
ESPL/04	M=L=5	18 ... 21 kHz 3.0 s	16 kHz	50 kHz, 50 kHz	$\lambda = 0$ / $\lambda = 3$

4. Lab task

4.1 Echo program

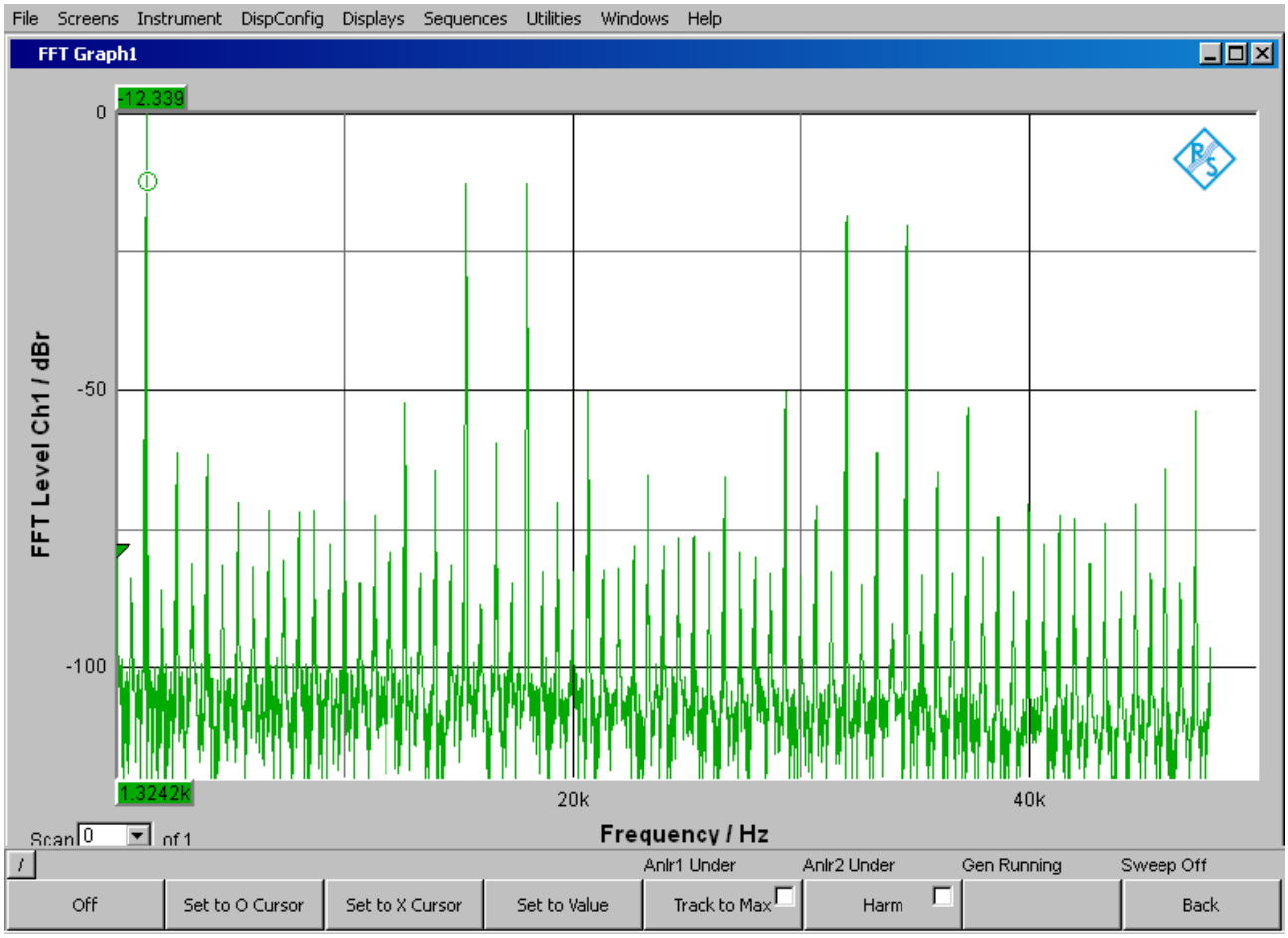
Take a screenshot of the **amplitude response** and attach it to the lab report. In the report, **describe** the shape of the frequency response and **explain** how this affects signal transmission in the DSP system.

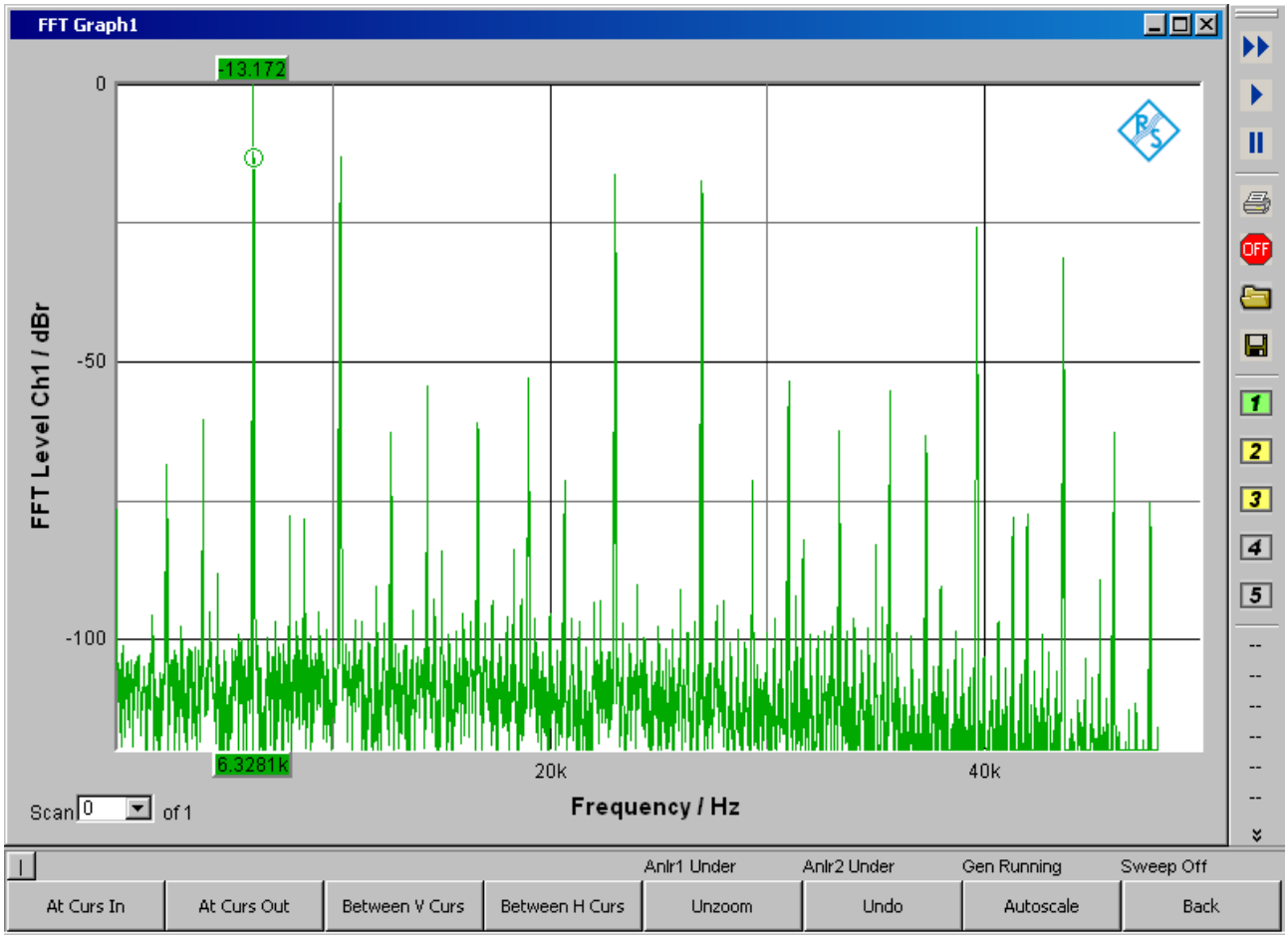


A low pass behavior can be seen at the amplitude response. Our Signals are in the range from 0 Hz to 21 kHz. This has a little effect that our signals are little bit attenuated in the higher frequency range.

4.2 Implementation of multirate system

After successful testing, insert the final C code of adcInt() and dacInt() into the lab report. Apply **a sine wave of frequency f_{bat}** as specified in Table 1 for your group to both inputs of the system and measure the **amplitude spectrum via FFT** at the output. Take a screenshot for the report. **Increase the frequency by 5 kHz** and compare the amplitude spectrum to the previous. Explain your observations.

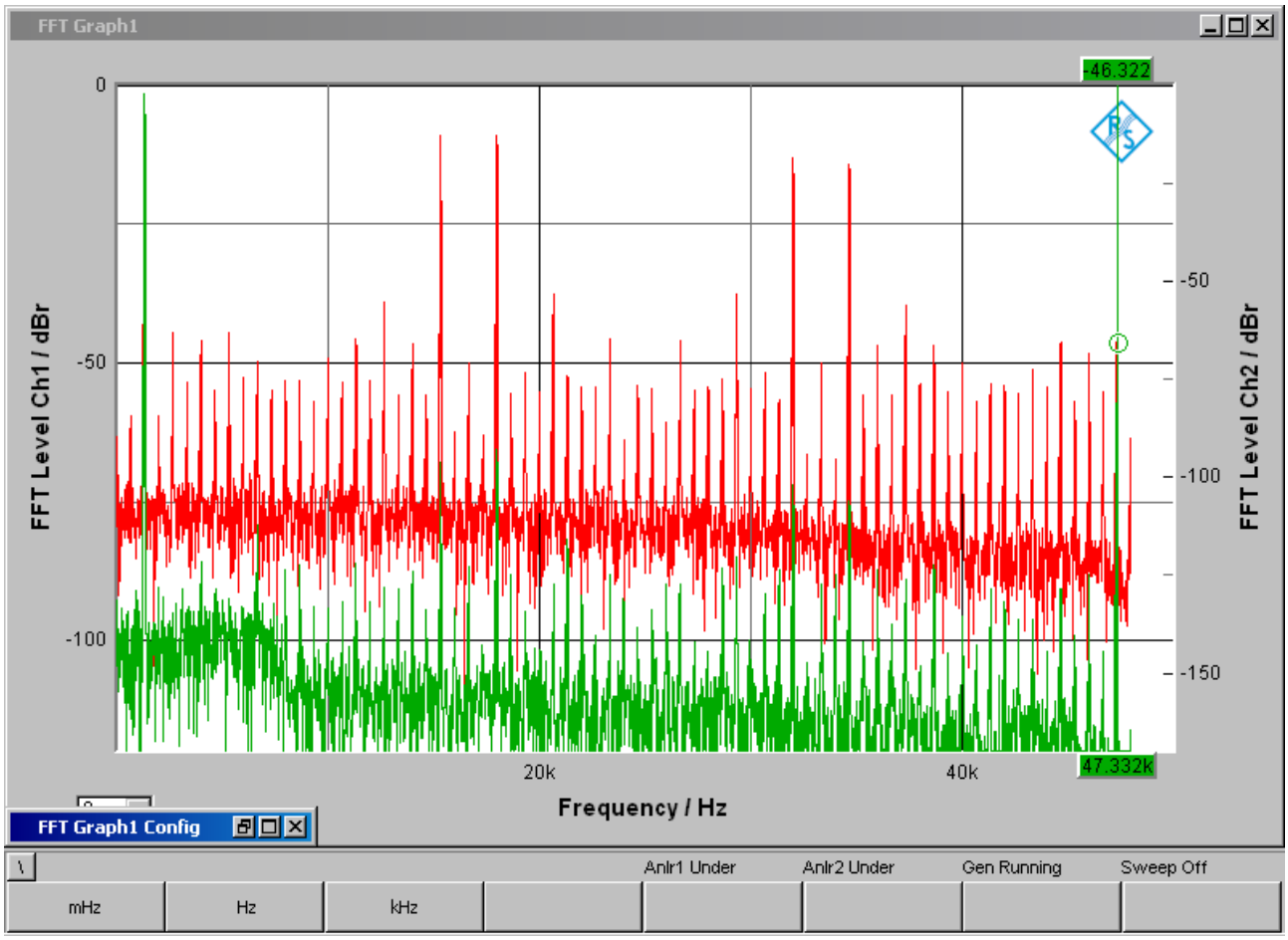

File Screens Instrument DispConfig Displays Sequences Utilities Windows Help
FFT Graph1
-12.339
0
-50
-100
FFT Level Ch1 / dBr
1.3242K
20k
40k
Frequency / Hz
Scan 0 of 1
Anlr1 Under
Anlr2 Under
Gen Running
Sweep Off
Off
Set to O Cursor
Set to X Cursor
Set to Value
Track to Max
Harm
Back

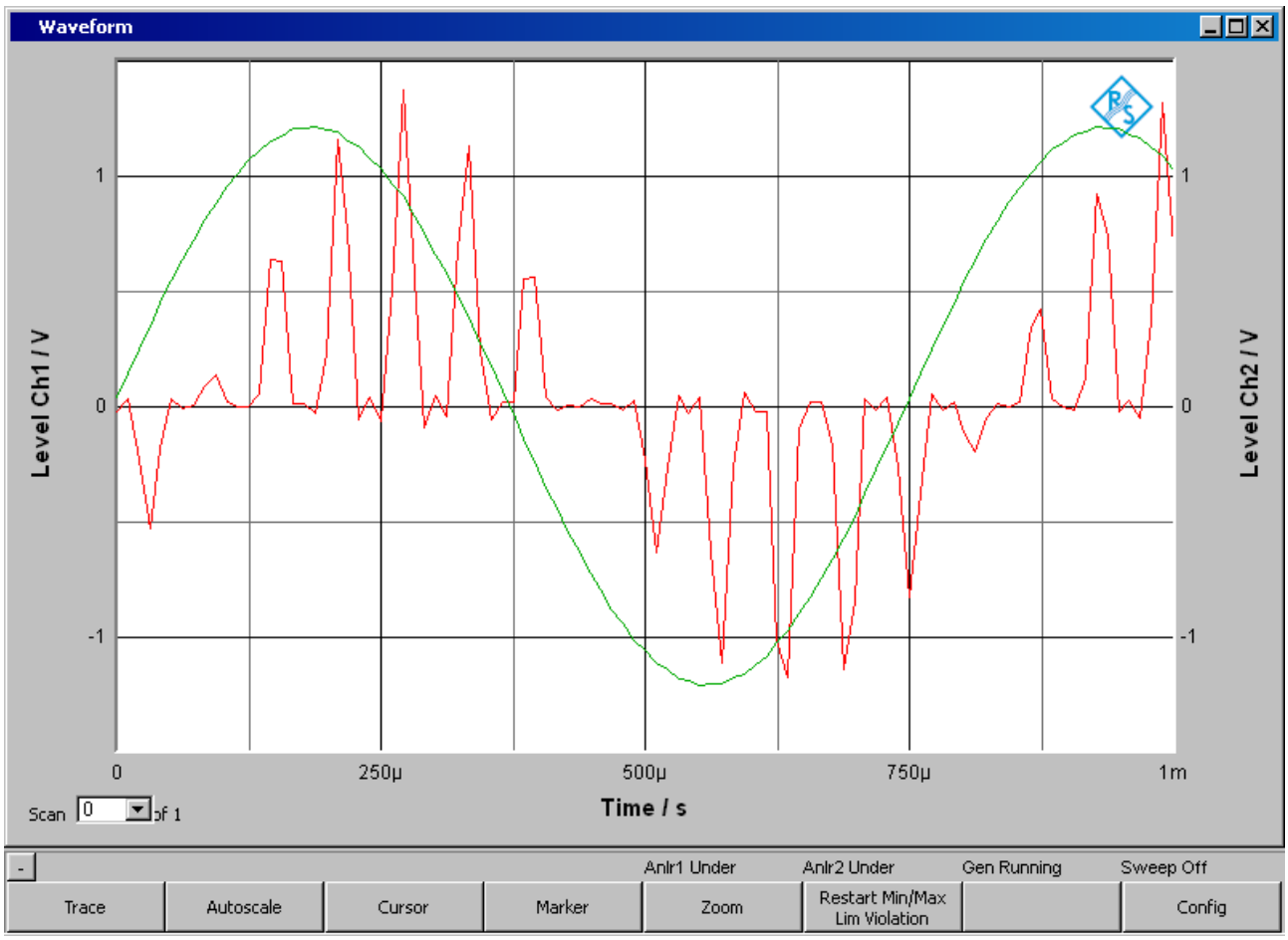

FFT Graph1
-13.172
0
-50
-100
FFT Level Ch1 / dBr
6.3281k
20k
40k
Frequency / Hz
Scan 0 of 1
Anlr1 Under
Anlr2 Under
Gen Running
Sweep Off
At Curs In
At Curs Out
Between V Curs
Between H Curs
Unzoom
Undo
Autoscale
Back

We can see at fbat = 18kHz our top plot and the expected peak at 1.3kHz and 15.33kHz as described in the preparation. And if we increase our bat frequency to 23kHz then the other peaks will be shifted 5kHz increased with the new sample frequency as before. (1.3kHz is now at 6.3kHz)

4.3 Insert the interpolation filter

Compare amplitude spectra and **time-domain signals** of both DAC channels 0 and 1 again for the bat frequency f_{bat} as given in the Table 1. Add relevant **screenshots** to your report and **explain** the results.





The red signal is the upsampled signal without the interpolation filter. It can be seen in the frequency plot that the low-pass filter is filtering the image spectra (the unwanted peaks at 15 and 18kHz) with 60dB attenuation and that signal is successfully interpolated.

4.4 Optional task: let the bats fly

A wave file containing bat sounds has been stored on the internship computer. These are not the original sounds, which are in the 40-50 kHz range, but were shifted to the upper human hearing range. Input the signal to your system and compare the animal sounds at input and output of the multirate system and with and without interpolation filters by listening and using the audio analyzer.

Congratulations! **You made it. Please submit with 15 min. after end of the session.**